MeshLens

Learn the pattern.
Reveal the exception.

Industrial mesh anomaly detection with a trained convolutional autoencoder, held-out evaluation and real model inference in the browser.

0.7594	19.30%	11/57
Held-out AUROC	Defect recall	Defects flagged

What was built

An audited dataset pipeline; original normal-only model development; separate calibration; frozen checkpoint and ONNX export; a browser inspection workspace; measured plots, tests, an executed notebook, and deployment instructions.

What the results mean

The fixed threshold flagged 11 defects and missed 46; it flagged 0 of 21 normal images. Precision is 100.00% and recall is 19.30%. These counts describe the chosen operating point on a small curated benchmark.

The score is a reconstruction-error statistic, not a defect probability. Residual maps support review and are not validated segmentation. This report summarizes this run's versioned artifacts.

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Technical report / reproducible portfolio release / 2026

01 / Problem and system design

Industrial grid material has repeated local structure. A broken or bent strand and foreign material can disturb that pattern. Defect-free images are easier to obtain than a comprehensive catalogue of every future defect.

MeshLens learns to reconstruct normal material. It exposes local reconstruction disagreement and compares an image-level error statistic with a frozen normal-calibrated threshold. A reviewer can examine both the reconstruction and residual evidence.

Stage	Implementation	Output
Acquire/audit	Pinned archive, safe extraction, hashes and image checks	Image-level manifest
Learn/select	Normal patches, normal tune and fixed synthetic alterations	Chosen from-scratch model
Calibrate/test	Separate normal threshold calibration; held-out real defects	Frozen rule and measured report
Inspect locally	Static app and ONNX Runtime Web	Score, reconstruction and evidence

Two environments, one frozen model

Python owns development, evaluation and export. The static web app loads the exported graph and executes it locally through WebAssembly. Visitors need no Python installation and send no image to a Python inference service.

User workflow

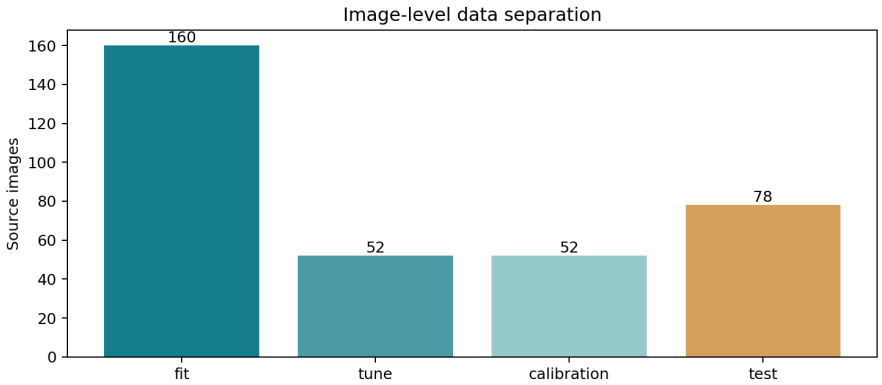
Choose an example, decode a local image, or sample a permitted camera frame; run inspection; compare input/reconstruction/residual views; review the score and threshold; optionally export local history.

The intended setting is a stable inspection view. New materials, lighting, scale or pose require representative camera validation.

02 / Dataset and data lineage

This acquisition records 342 real MVTec AD grid images and 57 defect masks. Source images are decoded and hashed; counts, duplicate checks and image-level roles are retained in the manifest.

Partition	Images	Use
Fit	160 normal	Optimize weights using normal patches
Tune	52 normal	Checkpoint/candidate/score selection with synthetic alterations
Calibration	52 normal	Freeze the normal-tail threshold
Official test	21 normal + 57 defects	Final real-image evaluation



Defect groups: bent 12, broken 12, glue 11, metal contamination 11 and thread 11. Patches stay in the source image's partition. Real defects/masks do not select weights, score aggregation or threshold.

Acquisition and rights

The archive is pinned to immutable mirror revision dc1c313de9de1be8e41fe6818dc43caac8601efa. Archive, image and mask hashes are recorded. The audit checks the pinned mirror and decoded structure; it does not independently compare every byte with a separate official download.

MVTec Software GmbH and original authors retain data rights. Images, masks, samples and dataset-derived material remain subject to CC BY-NC-SA 4.0; the original code's MIT license does not grant commercial data rights.

03 / Autoencoder and anomaly score

Every network parameter is trained from random initialization. The encoder compresses local texture and the decoder reconstructs RGB intensities. No pretrained feature extractor or external anomaly detector runs in the inference path.

Selected model: 25,339 parameters; input-noise standard deviation 0.035. Deployment uses the clean input directly.

Still PNG/JPEG/WebP inputs resolve EXIF orientation and composite alpha over opaque white before resizing. Bounds: 32-4096 pixels per side, at most 16 megapixels and 12 MiB. Browser/Pillow interpolation can differ; opaque benchmark tensors are unchanged.

Block	Operation	Contract
Input	RGB Float32 [0,1]	1x3x256x256
Boundary	Reflect-pad 16 pixels	Crop back after decoding
Encoder	3x3 convolution, then three stride-2 layers	3 > 12 > 16 > 24 > 8
Decoder	Three stride-2 transpose layers, RGB convolution	8 > 24 > 16 > 12 > 3
Activations	LeakyReLU hidden; linear bottleneck; sigmoid output	Output intensities [0,1]

Pixel residual

Square the input/reconstruction difference in each normalized RGB channel and average the three values for each pixel. No ground-truth mask enters this computation.

Image score

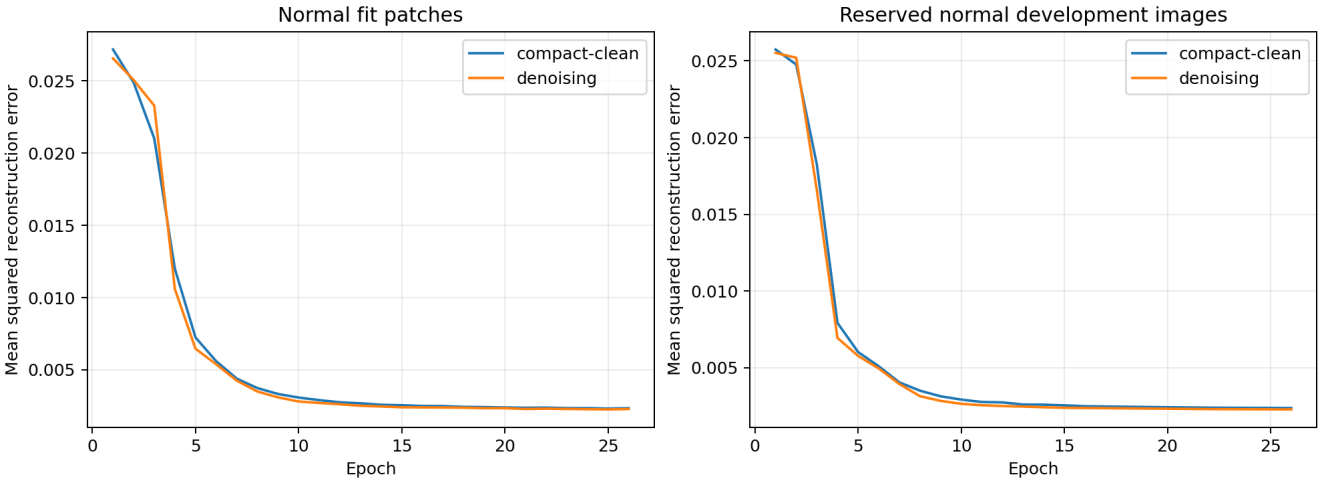
Average the largest 656 of 65,536 pixel residuals (1%, count rounded upward). This fixed aggregation emphasizes localized high-error regions.

Decision

Request review only when score > 0.02402208. Equality remains within the learned range. The score is not a calibrated probability, and its residual map is not validated segmentation.

04 / Training and selection

Training uses 64 x 64 random patches from normal fit images after image-level preprocessing. Horizontal/vertical flips vary normal patterns. Clean reconstruction and denoising candidates are compared; clean normal patches remain the targets.



Checkpoint selection minimizes full-image reconstruction MSE on reserved normal tune images. Candidate and aggregation selection use normal-versus-fixed-synthetic tune AUROC. Synthetic colored blocks, dark cuts and bright streaks are development proxies, not additional real test defects.

Candidate	Latent	Top pixels	Synthetic tune AUROC
compact-clean	4	1%	0.9094
compact-clean	4	5%	0.7792
denoising	8	1%	0.9124
denoising	8	5%	0.7885

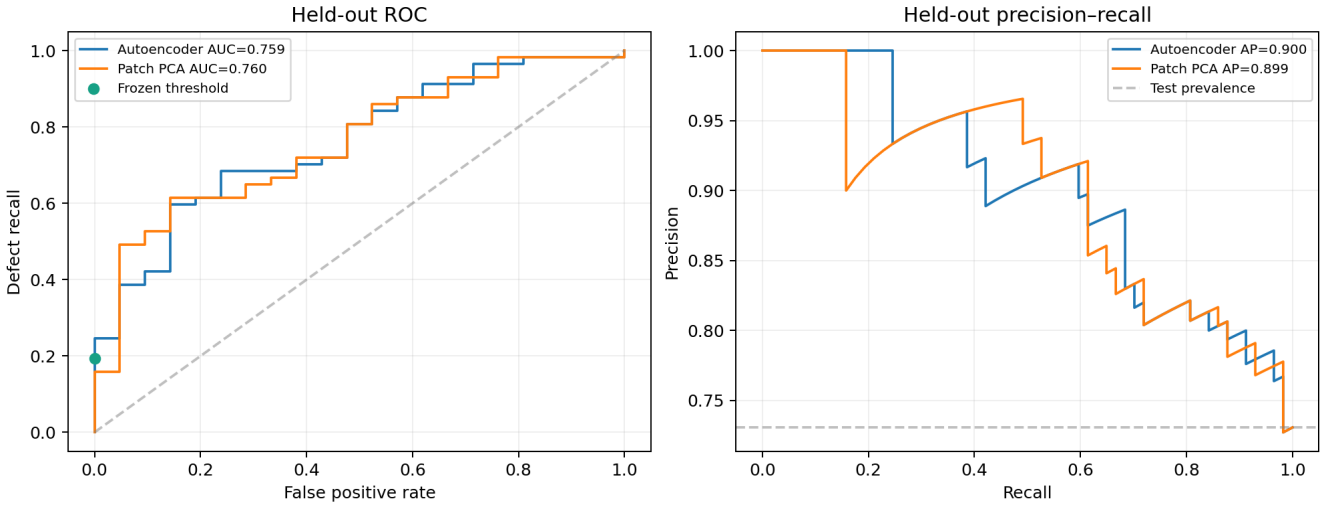
Frozen choice: denoising, 8 latent channels, epoch 26, top 1% residuals. Ties use the recorded lower-MSE/smaller-tail rule.

Separate normal calibration uses 52 images and alpha = 0.05. The finite-sample sorted rank ceil((52+1)*0.95)=51 targets a normal-tail operating point under exchangeability. It does not guarantee a false-alarm rate under image-formation shift.

05 / Held-out performance

The model, checkpoint, scoring rule and calibration threshold are fixed before official test scoring. Ranking metrics and fixed-threshold metrics describe different aspects of behavior.

Metric	Result	Meaning
AUROC	0.7594	Normal/defect score ranking
Average precision	0.9005	Precision-recall ranking
Precision	100.00%	Defects among flagged images
Recall	19.30%	Defects flagged for review
F1	32.35%	Precision/recall harmonic mean



Fixed-model stratified bootstrap 95% AUROC interval: 0.6391-0.8655. Resampling test images does not capture retraining or unseen-factory uncertainty.

Wilson 95% intervals: precision 74.12%-100.00%; recall 11.13%-31.34%; normal false alarms 0.00%-15.46%.

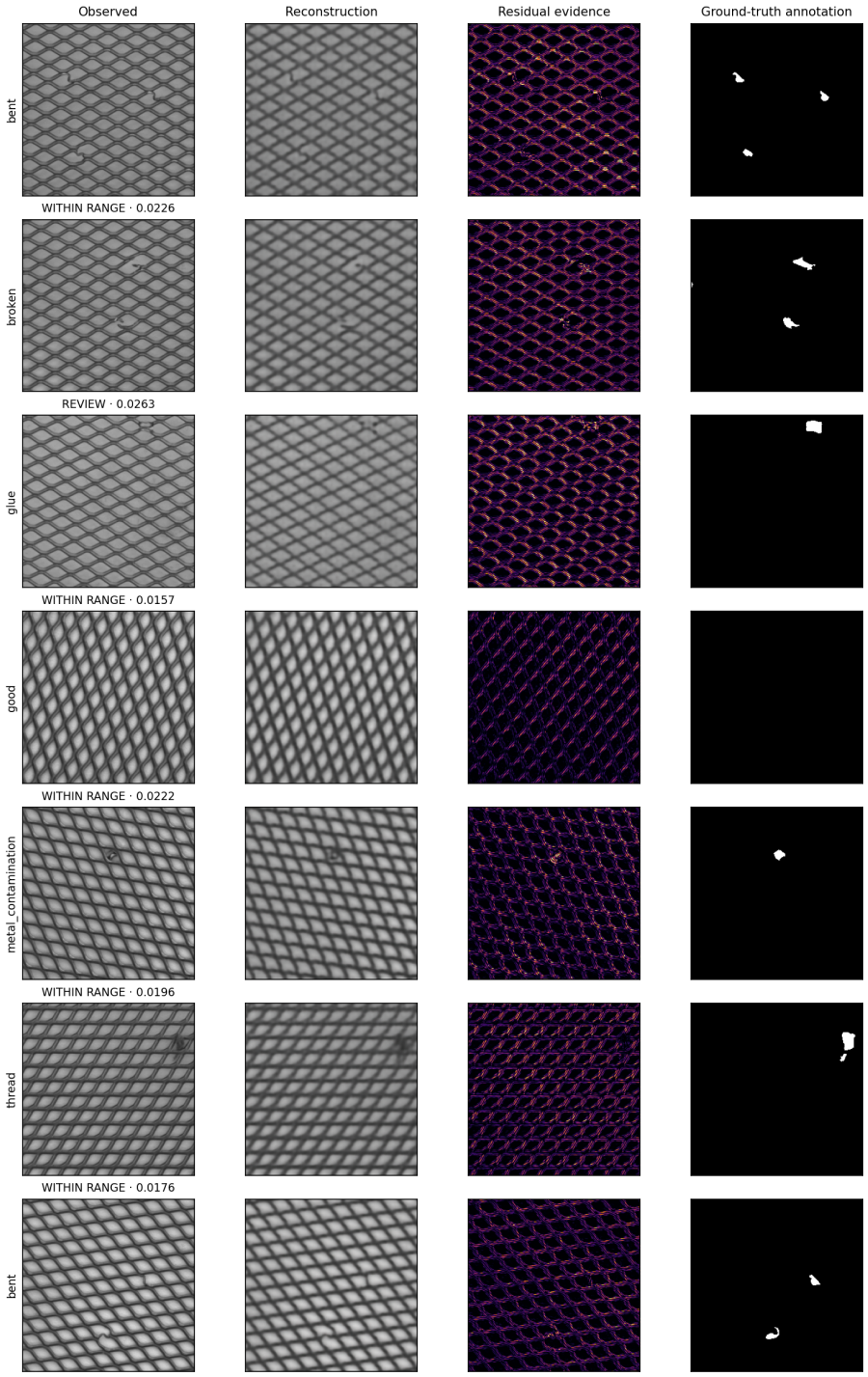
Patch PCA baseline: AUROC 0.7602, average precision 0.8987, recall 8.77%. Both methods retain their frozen calibration thresholds.

Operating point

The autoencoder flagged 11 defects and missed 46, with 0 false alarms among 21 normal images. Good observed precision can accompany low recall; zero observed false alarms does not imply zero population risk.

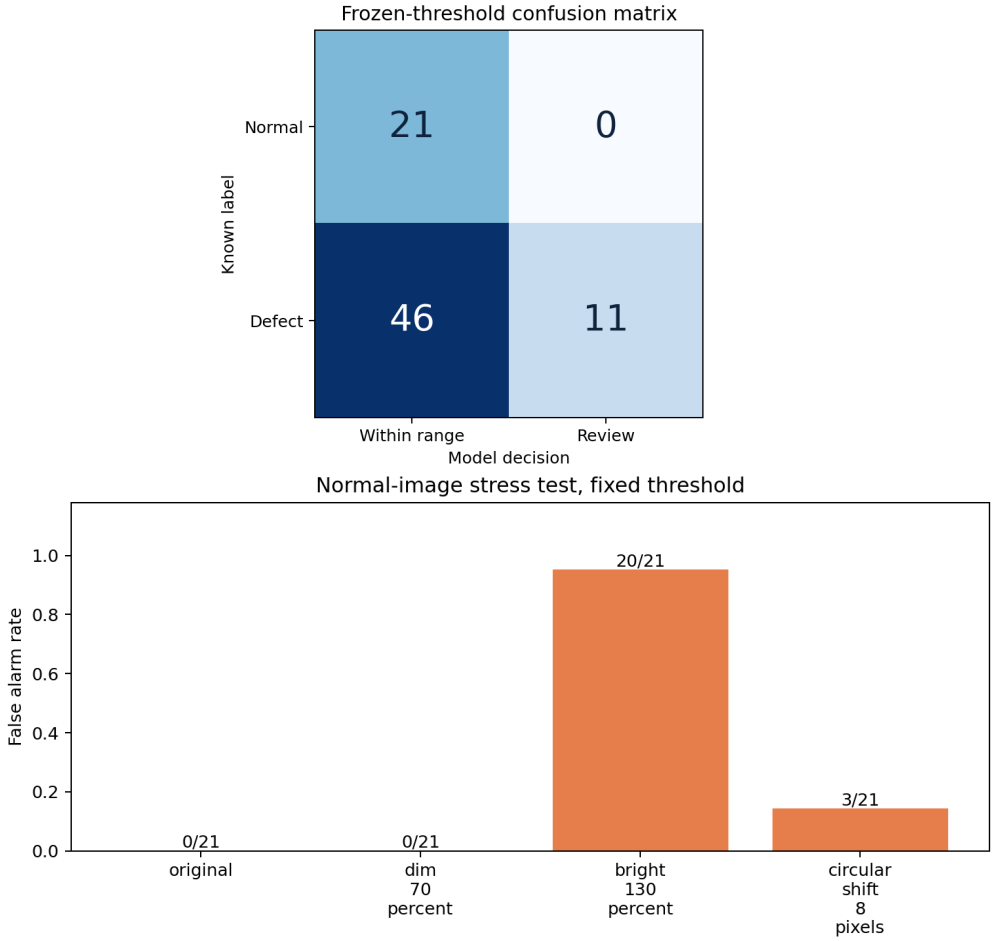
06 / Reconstruction evidence

The input, reconstruction and squared RGB residual expose why an image may receive a high score. A residual can also reflect normal texture phase, boundaries, contrast or distribution shift.



These are actual held-out model outputs. Ground-truth masks, if shown, support post-freeze analysis only and are unavailable during normal browser inference. A bright residual is not proof of a physical defect or validated pixel segmentation.

07 / Failure and robustness



Brightening normal images to 130% caused 20/21 false alarms; an 8-pixel circular shift caused 3/21. The model and threshold remained frozen.

These controlled image modifications probe sensitivity rather than estimate factory event rates. Held-out observations are reports, not a new threshold-tuning loop.

Deployment implication

Control lighting, view and material scale. Validate representative camera data and agree the false-alarm/missed-defect trade-off before operational use. This curated benchmark does not establish factory qualification or autonomous rejection reliability.

08 / Reproduction and verification

Versioned evidence includes acquisition hashes, the image-level split manifest, candidate records, epoch history, calibration values, per-image test predictions, export parity, plots and an executed notebook.

Rebuild and check

```
python -m pip install -r requirements-dev.lock
python -m pip install --no-deps -e .
python scripts/download_data.py
python scripts/train.py
python scripts/evaluate.py
python scripts/export_model.py
python -m pytest -q
npm ci
npm test
npm run build
```

Seeds and a bounded CPU schedule are recorded. Hardware/library differences can affect floating-point details; numerical agreement must be measured rather than assumed.

The release includes completed candidate checkpoints. The training command validates/reuses matching candidates and recomputes selection. For fresh optimization, first move candidate-*.pt and training-resume-*.pt out of artifacts; retain the release for comparison.

Verification scope

These are recorded cases, not proof that every possible edge case is covered. Software/mock-camera checks and physical hardware validation have separate scopes.

Source/model verification: passed; 145 Python tests, 145 passed, 0 failures, 0 errors and 0 skips.

Node.js browser-contract tests: 85 tests, 85 passed, 0 failures, 0 errors and 0 skipped.

Source-only checkout verification: passed; 145 Python tests, 144 passed, 0 failures, 0 errors and 1 skipped. The optional full raw-data audit is separate from mandatory tracked-model/example inference.

PyTorch/ONNX export check: 78 official test images, maximum tensor difference 2.98e-07, maximum score difference 4.29e-09, 0 changed decisions.

Actual ONNX Runtime Web WebAssembly under Node.js: passed; 78 shared canonical tensors, maximum score difference 3.01e-09, 0 changed decisions. Browser decoding, DOM interactions and physical cameras are outside this check.

Framework boundary diagnostic: PyTorch 2.9.0+cpu to 2.14.1+cpu on 130 calibration/test images. Strict decision parity failed; maximum score difference 3.52e-10. Changed decisions: calibration 1, official test 0. A calibration score exactly equal to the frozen threshold crosses it after a tiny floating-point change. Numerical tolerances are met, but exact decision equality is not. The threshold was not retuned. Browser/CLI inference use the frozen ONNX graph. No retraining is implied.

Walkthrough: 7/7 code cells executed with real outputs; 0 errors. Execution engine: socketless IPython.

09 / Deployment and scope

The website serves HTML, CSS, JavaScript, a local WebAssembly runtime, ONNX graph, metadata and licensed examples. Model inference happens in the browser, without a Python hosting process or remote inference API.

Concern	Behavior
Images	Decoded/inspected locally; not uploaded to a Python inference service
History	Bounded browser-local summaries; clear and CSV export
Camera	User initiated; serial sampled frames; browser permission
Model	Frozen graph/metadata; no website-startup training
Hosting	Static dist output; documented local/static-host paths
Audience	Live publication and access controls verified separately

Industrial audit conclusion

This checkpoint flagged 11/57 defects and missed 46. It is an industrial-mesh prototype for human review and does not qualify automatic acceptance/rejection. New materials, optics, motion and line speeds need independent representative validation.

Camera mode samples serially, waiting 250/1000/2500 ms after completed inspections (default 1000 ms). It is not full-video-FPS processing. Physical camera hardware, target-line latency and long-duration throughput remain unverified. Model-call time alone is not end-to-end latency.

The detailed project audit (reports/PROJECT_AUDIT.md) separates fixes, passing checks, open limitations and physical-unverified scenarios. A finite test suite cannot prove the absence of undiscovered bugs.

Privacy and persistence

Local history can be cleared or altered by the browser user, does not synchronize across devices and is not an authenticated production audit trail. Hosting audience is configured independently of model training.

Named-scope dependency advisory scans: recommended runtime: passed, 36/36 packages, 0 findings; development: passed, 97/97 packages, 0 findings; historical original-training lock: 5 findings (provenance only, not the default install); npm: 0 findings. These scans are not security certification.

Rights and authorship

Original MeshLens code/documentation: Kumar Waibhav Akshat, MIT. MVTec AD images and masks: MVTec Software GmbH and original authors, CC BY-NC-SA 4.0. Review the dataset card and third-party notices before redistribution.

The delivered machine-readable experiment records, test outputs, notebook and deployment record are the source of truth for this release.